

Notes on Adjusted R Squared

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ADJUSTED R SQUARED

Adjusted R Squared is a modified version of the R Squared metric that takes into account the number of predictors in a regression model.

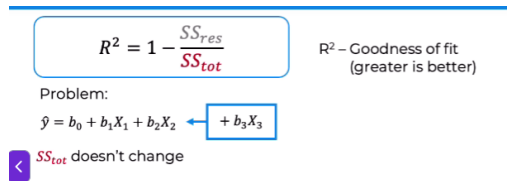


Fig. 1. Illustration of Adjusted R^2 .

It is calculated using the formula:

$$\text{Adjusted } R^2 = 1 - \left(\frac{(1 - R^2)(n - 1)}{n - k - 1} \right)$$

where:

- R^2 is the coefficient of determination (R Squared),
- n is the number of observations,
- k is the number of predictors in the model.

The Adjusted R Squared value can be negative if the model fits the data very poorly. Unlike regular R^2 , which always increases when additional predictors are added, Adjusted R^2 penalizes the inclusion of unnecessary variables.

A higher Adjusted R^2 generally indicates a better-fitting model, while a lower value suggests that the predictors do not explain the variation in the response variable effectively. This makes Adjusted R^2 useful for comparing regression models that contain different numbers of predictors.

However, Adjusted R^2 should not be the only criterion used for model selection. Statistical significance tests, residual analysis, and domain-specific considerations should also be taken into account when evaluating model performance.

If SS_{tot} doesn't change, then the Adjusted R Squared will decrease as more predictors are added, which is a key reason why it is preferred over regular R Squared for model comparison. If SS_{res} will decrease or stay the same as more predictors are added, but the Adjusted R Squared will only increase if the new predictor improves the model sufficiently to offset the penalty for adding more predictors. (This is because of Ordinary Least Squares: $SS_{res} \rightarrow \min$)